Essential in Fluent (40 Hours)

S.No	Session	Topics	Type	Duration (Hours)
1	Session 1	**INTRODUCTION TO CFD** CFD theory Fundamentals and applications ANSYS Fluent application Fluent simulation process	Theory/Demo	2.0 Hrs
2	Session 2	Exercises included for practice	Practicals	2.0 Hrs
3	Session 3	**ANSYS FLUENT GUI AND SOFTWARE PRELIMINARIES** Solver fundamentals Boundary conditions CFD solving workflow Geometry in Design Modeler	Theory/Demo	2.0 Hrs
4	Session 4	Exercises included for practice	Practicals	2.0 Hrs
5	Session 5	**FLOW MIX AND HEAT TRANSFER (3D)** Meshing using ANSYS Meshing application CFD Simulation set up: Models, Defining regions, Material definition, Cell Zone conditions Extracting preliminary results Coupled solver Mesh Adaption	Theory/Demo	2.0 Hrs
6	Session 6	Exercises included for practice	Practicals	2.0 Hrs
7	Session 7	**TRANSONIC FLOW – EXTERNALLY COMPRESSIBLE** CFD Simulation set up: Models, Defining regions, Material definition Solution Post Processing	Theory/Demo	2.0 Hrs
8	Session 8	Exercises included for practice	Practicals	2.0 Hrs
9	Session 9	**MODELING MULTI SPECIES FLOW** 1 Case Study Simulation boundary conditions Set solution controls Calculation Results	Theory/Demo	2.0 Hrs
10	Session 10	Exercises included for practice	Practicals	2.0 Hrs
11	Session 11	**TURBULENCE MODEL IN FLUENT** Setting up model Materials and operating conditions Solution extraction Convergence modeling Verification and validation	Theory/Demo	2.0 Hrs
12	Session 12	Exercises included for practice	Practicals	2.0 Hrs
13	Session 13	**MODELING PERIODIC HEAT FLOW** Creation of zones Model setup Material definition Cell zone definitions Periodicity definition Boundary conditions Solution extraction Advanced post processing	Theory/Demo	2.0 Hrs
14	Session 14	Exercises included for practice	Practicals	2.0 Hrs
15	Session 15	**RADIATION AND CONVECTION** Model setup Thermal boundary conditions Solution Post Processing Comparing the contour plots S2S definition and solving	Theory/Demo	2.0 Hrs
16	Session 16	Exercises included for practice	Practicals	2.0 Hrs
17	Session 17	**TURBULENT FLOW IN HEAT EXCHANGER** Model Setup Boundary conditions Solving Post Processing	Theory/Demo	2.0 Hrs
18	Session 18	Exercises included for practice	Practicals	2.0 Hrs
19	Session 19	**SIPHONING USING MULTIPHASE** Model setup Problem setup Initialize the flow field Generate animation	Theory/Demo	2.0 Hrs
20	Session 20	Exercises included for practice	Practicals	2.0 Hrs